Geographic and Socioeconomic Bias in Satellite-Based Flood Detection Datasets: Implications for Equitable Disaster Response

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Abstract

Satellite-based flood detection using machine learning has become critical for rapid disaster response and resource allocation during flooding events. However, current flood detection datasets exhibit significant geographic and socioeconomic bias, with training data predominantly sourced from well-monitored, high-income regions such as the United States, Europe, and East Asia. This concentration of data in affluent regions creates systematic performance disparities when models are deployed in underrepresented areas, many of which face the highest flood vulnerability due to climate change and limited infrastructure. This research proposes an experimental investigation into the extent and impact of geographic bias in flood detection datasets, hypothesizing that models trained on geographically imbalanced data will demonstrate degraded performance in underrepresented regions during actual flood events. Through systematic analysis of existing datasets, evaluation of state-of-the-art models across diverse

geographic contexts, and design of targeted data collection strategies, this work aims to quantify these disparities and propose methodologies for more equitable flood monitoring systems. The outcomes of this research have direct implications for climate justice, as they will inform strategies to ensure AI-powered disaster response systems serve all vulnerable populations effectively, regardless of geographic location or economic status.

1 Introduction

Flooding represents one of the most destructive and frequently occurring natural disasters globally, affecting millions of people annually and causing billions of dollars in economic damage [1]. Climate change is intensifying the frequency, severity, and unpredictability of flood events worldwide, with projections indicating that flood risk will increase substantially in the coming decades, particularly in vulnerable regions across South Asia, Sub-Saharan Africa, and small island developing states [2]. In this context, the development of rapid, accurate flood detection and mapping systems has become critical for effective disaster response, emergency resource allocation, and post-disaster recovery planning.

Satellite remote sensing has emerged as a powerful tool for flood monitoring, offering the ability to rapidly assess large geographic areas that may be inaccessible to ground-based observers during active flooding events [3]. Recent advances in machine learning and computer vision have enabled automated flood detection from satellite imagery, with deep learning models achieving impressive performance in delineating flood extent from synthetic aperture radar (SAR) and optical satellite data [4]. These AI-powered systems promise to accelerate disaster response by providing near-real-time flood maps to emergency managers, humanitarian organizations, and affected communities.

However, the effectiveness of machine learning models is fundamentally dependent on the quality, diversity, and representativeness of their training data [5]. An emerging body of research in fairness and bias in AI systems has demonstrated that geographic and demographic imbalances in training datasets can lead to systematic performance disparities, with models exhibiting reduced accuracy when applied to populations or contexts underrepresented in their training data [6, 7]. In the domain of satellite-based Earth observation, these concerns are particularly acute, as data collection, annotation, and validation efforts have historically concentrated in well-resourced regions with established research infrastructure and government monitoring programs.

1.1 Problem Statement

This research investigates the following central question: **Do current satellite-based flood detection datasets exhibit geographic and socioeconomic bias, and does this bias lead to degraded model performance in underrepresented regions during actual flood events?**

Preliminary examination of widely-used flood detection datasets reveals concerning patterns of geographic concentration. The Sen1Floods11 dataset, one of the most comprehensive publicly available datasets for training flood detection models, contains 4,831 image chips covering flood events, yet the geographic distribution of these events is heavily skewed toward North America, Europe, and East Asia [8]. Similarly, the FloodNet dataset, designed for high-resolution flood damage assessment, primarily includes imagery from the United States [9]. The MODIS-based flood products and Copernicus Emergency Management Service flood maps, while more globally distributed, show disproportionate validation and ground-truth collection in high-income countries [10].

This geographic imbalance is particularly troubling because the regions most underrepresented in training data are often those facing the highest flood vulnerability. According to the United Nations Office for Disaster Risk Reduction, approximately 90% of disaster-related deaths occur in low- and middle-income countries, with flooding being a leading cause [11]. Countries in South Asia, Sub-Saharan Africa, and Southeast Asia experience frequent devastating floods but lack the satellite monitoring infrastructure, ground validation networks, and labeled datasets that characterize better-resourced regions.

The hypothesis underlying this research is that machine learning models trained on geographically biased datasets will exhibit performance degradation when applied to flood detection in underrepresented regions. This degradation may manifest as reduced detection accuracy, increased false positive rates, or decreased sensitivity to flood extent in these critical areas. Such disparities would have profound implications for disaster response equity, potentially resulting in delayed emergency response, misallocation of humanitarian resources, and ultimately, preventable loss of life in the communities most vulnerable to climate change impacts.

1.2 Motivation

The motivation for this research stems from both scientific and ethical imperatives. From a scientific perspective, understanding the geographic biases in flood detection datasets is essential for improving model robustness and generalization. Current evaluation practices often rely on test sets drawn from the same geographic distributions as training data, potentially masking significant performance gaps that emerge when models are deployed in novel contexts [7]. By systematically quantifying these biases and their impacts, this research will contribute to the broader understanding of dataset representativeness in Earth observation applications and inform best practices for more robust model development.

From an ethical standpoint, the stakes are considerably higher. Flood detection systems are not academic exercises—they are deployed in life-or-death situations where accurate, timely information directly influences emergency response decisions. When AI systems exhibit geographic performance disparities, they effectively encode and amplify existing global inequalities. Communities in low-income countries, which have contributed least to climate change yet face its most severe impacts, are doubly disadvantaged: first by increased flood risk, and second by AI systems that may be less reliable in their regions due to training data imbalances.

This research is aligned with growing calls for climate justice and equitable AI systems. The benefits of solving this problem include: (1) improved disaster response through more accurate flood detection across all regions, (2) equitable resource allocation ensuring all populations receive appropriate assistance, (3) better climate adaptation planning with accurate historical flood mapping, (4) scientific advancement in developing robust Earth observation models, and (5) methodological innovation in detecting and mitigating geographic bias in satellite-based datasets.

1.3 Research Approach

This research will proceed through several integrated phases of experimental investigation. First, a comprehensive analysis of existing flood detection datasets will quantify geographic distribution, examining the spatial coverage, temporal distribution, and socioeconomic characteristics of regions represented in training data. Second, state-of-the-art flood detection models will be evaluated on carefully constructed test sets that include both well-represented and underrepresented regions. Third, targeted data collection from underrepresented regions will be proposed and, where feasible, executed. Finally, the research will assess whether incorporating data from underrepresented regions into training sets leads to improved model performance across all geographic contexts.

1.4 Expected Contributions

This research will contribute to both the scientific understanding of geographic bias in machine learning systems and the practical development of more equitable flood monitoring technologies. Specific expected contributions include: (1) quantitative characterization of geographic and socioeconomic bias in existing flood detection datasets, (2) empirical evidence of performance disparities across geographic regions, (3) methodological frameworks for detecting and measuring geographic bias in Earth observation datasets, (4) targeted data collection strategies to address gaps in underrepresented regions, and (5) recommendations for dataset development and deployment practices that promote geographic equity in AI-powered disaster response systems.

2 Background and Related Work

This section reviews the literature on flood detection methods, datasets, and the broader issues of geographic bias in machine learning systems. The review identifies gaps in existing work and establishes the feasibility and necessity of the proposed research.

2.1 Literature Search Methodology

A systematic literature search was conducted using combinations of keywords including: *flood detection, flood mapping, satellite imagery, remote sensing, SAR, deep learning, geographic bias, dataset bias, fairness, disaster response*. The search employed Google Scholar, IEEE Xplore, and Remote Sensing journals, with inclusion criteria focusing on peer-reviewed publications from 2015-2024 addressing flood detection datasets, methods, or bias in Earth observation systems.

2.2 Flood Detection Datasets

2.2.1 Sen1Floods11

Bonafilia et al. [8] introduced Sen1Floods11, containing 4,831 georeferenced image chips from 11 flood events using Sentinel-1 SAR imagery. The dataset provides hand-labeled flood masks and represents a major benchmark for training flood detection models.

Strengths: Sen1Floods11 is one of the first large-scale public datasets for SAR-based flood detection. SAR's ability to penetrate clouds makes it invaluable for operational disaster response. The dataset includes diverse flood types (riverine, coastal, flash floods).

Limitations: Despite multi-continental coverage, the dataset exhibits strong geographic concentration in North America, Europe, and East Asia. Africa and South America are significantly underrepresented. With only 11 discrete flood events, the dataset may not capture the full diversity of global flooding scenarios. Annotation relied on expert labelers from high-income country institutions, potentially introducing subtle biases. Ground-truth validation was opportunistic, favoring well-monitored regions.

2.2.2 FloodNet

Rahnemoonfar et al. [9] presented FloodNet, a high-resolution aerial imagery dataset for post-flood damage assessment in the United States, with semantic labels for buildings, roads, water, and damaged infrastructure.

Strengths: FloodNet addresses critical disaster response needs with high spatial resolution enabling detailed damage analysis. Multiple semantic categories support emergency response prioritization.

Limitations: The dataset is geographically restricted almost entirely to the United States, particularly Gulf Coast and East Coast hurricane events. It lacks representation of informal settlements, rural developing country contexts, or different construction practices. Flood types are limited to tropical cyclone events, missing riverine and monsoon flooding common in South Asia.

2.2.3 Global Flood Database

Tellman et al. [10] developed a global flood database using 3,000+ satellite images spanning 2000-2018, revealing that population exposed to floods increased 24% over this period, with disproportionate impacts in South and Southeast Asia.

Strengths: The temporal span enables long-term trend analysis and documents increasing global flood exposure, particularly in vulnerable regions.

Limitations: The authors acknowledge that flood detection accuracy varies substantially by region due to differences in satellite coverage, cloud contamination, and validation data availability. Regions with established hydrological monitoring networks have better-validated flood extents, creating circular validation problems in unmonitored regions.

2.3 Machine Learning Methods for Flood Detection

2.3.1 Deep Learning Approaches

Nemni et al. [4] developed fully convolutional neural networks for rapid flood segmentation in SAR imagery, achieving high accuracy on Sen1Floods11 using U-Net architecture. Sarker et al. [12] proposed deep learning combining CNN features with domain knowledge about flood behavior, showing improved performance on complex urban flooding scenarios.

Limitations: These models inherit geographic biases from their training datasets. Nemni et al. acknowledge performance may degrade in regions with environmental characteristics dissimilar to training data. Transfer learning to new geographic regions was not comprehensively evaluated. Sarker's incorporation of domain knowledge may encode assumptions valid for some regions but not others.

2.3.2 Transfer Learning and Domain Adaptation

Boni et al. [13] investigated transfer learning for flood detection, training models on data-rich regions and adapting them to data-scarce regions. They found that transfer learning success depends heavily on similarity between source and target domains—when environmental characteristics differ substantially (tropical vs. temperate, mountainous vs. flat), simple transfer learning provides limited benefit.

Limitations: More sophisticated domain adaptation techniques are needed but have not been systematically evaluated across diverse geographic contexts for flood detection.

2.4 Geographic Bias in Machine Learning

2.4.1 Dataset Bias in Computer Vision

Torralba and Efros [5] provided seminal analysis demonstrating that models trained on one dataset often perform poorly on others due to subtle differences in image characteristics and photographer biases. Their work established that cross-dataset generalization is a fundamental challenge in computer vision.

Implications: If general object recognition datasets exhibit strong biases despite efforts to be comprehensive, specialized Earth observation datasets with smaller communities and limited resources likely exhibit even stronger geographic biases.

2.4.2 Geographic Representation

Shankar et al. [7] demonstrated that datasets like ImageNet severely underrepresent developing countries, with object recognition models performing 5-10 percentage points worse on images from underrepresented regions. De Vries et al. [14] showed these geographic imbalances persist across many computer vision benchmarks and require deliberate, targeted data collection to address.

Implications: The patterns identified—concentration in wealthy regions, performance degradation in underrepresented areas—directly parallel concerns in flood detection. However, stakes are higher in flood detection where performance disparities affect life-or-death disaster response decisions.

2.4.3 Bias in Earth Observation

Patel et al. [15] analyzed land cover classification datasets, finding systematic underrepresentation of certain ecosystem types and regions, with resulting models performing poorly on underrepresented classes. Rolf et al. [16] examined poverty prediction from satellite imagery, revealing models trained on certain countries perform poorly when deployed elsewhere, highlighting how geographic bias perpetuates inequalities in resource allocation systems.

Implications: Land cover classification shares methodological similarities with flood detection (pixel-wise segmentation of satellite imagery), suggesting similar biases likely exist in flood detection datasets. Rolf et al.'s work on poverty mapping is particularly relevant as it addresses applications where algorithmic failures directly harm vulnerable populations.

2.5 Fairness in Disaster Response AI

Buolamwini and Gebru [6] demonstrated significant accuracy disparities across demographic groups in commercial computer vision systems. Their methodological approach—evaluating performance disparities across subgroups and tracing them to training data imbalances—provides a template for analyzing geographic disparities in flood detection.

Crawford and Finn [17] critically examined AI in disaster response, arguing that these technologies can reinforce existing inequalities if not carefully designed. They note that disaster response technologies often privilege regions with better data infrastructure, potentially directing resources toward already better-served areas. Luers et al. [18] found that AI-powered climate adaptation systems are predominantly deployed in wealthy regions, leaving the most climate-vulnerable populations without access to these tools.

Implications: This creates a troubling pattern where communities most harmed by climate change are least likely to benefit from AI technologies. Flood detection systems risk following this pattern if geographic biases are not explicitly addressed.

2.6 Feasibility and Gaps in Existing Work

2.6.1 Room for Improvement

Despite advances in flood detection technology, no study has comprehensively quantified geographic bias in flood detection datasets or systematically evaluated performance disparities across regions. Existing work focuses primarily on technical performance in well-monitored areas, leaving critical gaps in understanding global system performance.

2.6.2 Data Availability

Multiple flood detection datasets are publicly available (Sen1Floods11, FloodNet, MODIS products, Copernicus services), enabling dataset bias analysis. Sentinel-1 and Sentinel-2 satellites provide global coverage, allowing performance evaluation across diverse regions even where labeled training data is limited. Historical flood event databases (DFO, GDIS) document floods globally, enabling identification of events in underrepresented regions for targeted study.

2.6.3 Methodological Foundations

Literature on dataset bias [5,7] and fairness in ML [6] provides established frameworks and metrics adaptable to flood detection. Geospatial analysis tools and statistical methods for quantifying geographic distributions are mature and accessible.

2.6.4 Summary of Limitations

Existing work exhibits several critical limitations: (1) nearly all flood detection datasets show strong geographic concentration in wealthy regions, (2) most studies evaluate models on test sets from similar geographic distributions as training data, potentially masking performance disparities, (3) ground-truth validation relies on infrastructure unevenly distributed globally, (4) few studies explicitly examine geographic bias in flood detection or quantify impacts on disaster response equity, and (5) limited practical strategies exist for creating geographically representative datasets or developing globally generalizable models.

The proposed research directly addresses these gaps by comprehensively quantifying geographic bias in flood detection datasets, evaluating model performance across geographic strata including underrepresented regions, and proposing methodologies for more equitable dataset development.

3 Methodology

This section presents the proposed methodology for investigating geographic bias in flood detection datasets and evaluating its impact on model performance. The approach consists of four integrated components: (1) geographic bias quantification in existing datasets, (2) cross-regional model evaluation, (3) bias-aware data augmentation, and (4) equitable model development.

3.1 Overview of Proposed Approach

Our methodology addresses the research question through a systematic, multi-phase investigation. Figure 1 illustrates the overall workflow.

```
Phase 1: Dataset Analysis
    -> Geographic Distribution Metrics
    -> Socioeconomic Stratification
    -> Bias Quantification

Phase 2: Cross-Regional Evaluation
    -> Baseline Model Selection
    -> Stratified Test Set Construction
    -> Performance Gap Analysis

Phase 3: Bias-Aware Augmentation
    -> Targeted Data Collection
    -> Synthetic Minority Oversampling
    -> Domain Adaptation

Phase 4: Equitable Model Development
    -> Geographically-Balanced Training
    -> Fairness-Constrained Optimization
    -> Multi-Regional Validation
```

Figure 1: Proposed methodology workflow for investigating geographic bias in flood detection systems.

3.2 Phase 1: Geographic Bias Quantification

3.2.1 Mathematical Framework

Let $\mathcal{D} = \{(x_i, y_i, g_i)\}_{i=1}^{N}$ represent a flood detection dataset where $x_i \in \mathbb{R}^{H \times W \times C}$ is a satellite image, $y_i \in \{0, 1\}^{H \times W}$ is the corresponding binary flood mask, and $g_i \in \mathcal{G}$ represents the geographic region where the image was captured.

We define the **Geographic Representation Index (GRI)** for region g as:

$$\text{GRI}(g) = \frac{N_g/N}{A_g/A_{total}} \times \frac{F_g/F_{total}}{P_g/P_{total}} \tag{1}$$

where:

- N_g = number of samples from region g
- N = total dataset size
- A_g = land area of region g
- A_{total} = total global land area
- F_g = historical flood frequency in region g
- F_{total} = total global flood events
- P_g = population at flood risk in region g
- P_{total} = global population at flood risk

A GRI value of 1.0 indicates proportional representation relative to geographic area, flood frequency, and population exposure. GRI > 1.0 indicates overrepresentation; GRI < 1.0 indicates underrepresentation.

3.2.2 Socioeconomic Stratification

We stratify regions by GDP per capita and compute the **Socioeconomic Disparity Metric (SDM)**:

$$\text{SDM} = \frac{1}{|\mathcal{G}|} \sum_{g \in \mathcal{G}} \left| \frac{N_g/N}{\text{GDP}_g / \sum_{g'} \text{GDP}_{g'}} - 1 \right| \tag{2}$$

Higher SDM values indicate greater correlation between dataset representation and economic wealth, suggesting socioeconomic bias.

3.2.3 Implementation Algorithm

Algorithm 1 Geographic Bias Quantification

Input: Dataset $\mathcal{D}$, region definitions $\mathcal{G}$, metadata (area, flood frequency, GDP)
Output: GRI scores, SDM, bias report

// Step 1: Extract geographic metadata
for each sample (x_i, y_i, g_i) in $\mathcal{D}$ **do**
  Extract coordinates from image metadata
  Assign to region $g_i \in \mathcal{G}$ based on coordinates
  Record flood event characteristics
end for

// Step 2: Compute representation metrics
for each region $g \in \mathcal{G}$ **do**
  Compute N_g (sample count)
  Retrieve A_g, F_g, P_g from external databases
  Calculate GRI(g) using Equation (1)
end for

// Step 3: Socioeconomic analysis
Stratify regions by GDP quintiles
Compute SDM using Equation (2)

// Step 4: Visualize and report
Generate geographic heat maps showing GRI distribution
Create scatter plots: GDP vs. representation
Output statistical significance tests (Spearman correlation)

3.3 Phase 2: Cross-Regional Model Evaluation

3.3.1 Stratified Test Set Construction

To evaluate performance disparities, we construct geographically stratified test sets that ensure representation across diverse regions. Define:

$$\mathcal{T} = \bigcup_{g \in \mathcal{G}} \mathcal{T}_g, \quad |\mathcal{T}_g| = k \quad \forall g \tag{3}$$

where $\mathcal{T}_g$ contains k samples from region g, ensuring balanced evaluation across all geographic strata.

3.3.2 Performance Disparity Metrics

For a trained model $f : \mathbb{R}^{H \times W \times C} \rightarrow [0,1]^{H \times W}$, we measure region-specific performance using Intersection over Union (IoU):

$$\text{IoU}_g = \frac{1}{|\mathcal{T}_g|} \sum_{(x,y) \in \mathcal{T}_g} \frac{|f(x) \cap y|}{|f(x) \cup y|} \tag{4}$$

The **Performance Disparity Index (PDI)** quantifies inequality across regions:

$$\text{PDI} = \frac{\max_g \text{IoU}_g - \min_g \text{IoU}_g}{\text{mean}_g \text{IoU}_g} \tag{5}$$

Lower PDI indicates more equitable performance. We also compute the Gini coefficient of IoU scores across regions to measure inequality.

3.3.3 Statistical Significance Testing

To determine whether performance disparities are statistically significant, we employ:

- **Kruskal-Wallis H-test** to test if median IoU differs significantly across geographic regions
- **Dunn's post-hoc test** with Bonferroni correction for pairwise regional comparisons

- **Linear mixed-effects models** to control for confounding factors (flood type, image quality, temporal factors)

3.4 Phase 3: Bias-Aware Data Augmentation

3.4.1 Targeted Data Collection Strategy

Based on Phase 1 analysis, we identify underrepresented regions with GRI$(g) < 0.5$. For these regions, we propose targeted data collection:

$$N_g^{\text{target}} = N \times \frac{A_g}{A_{total}} \times \frac{F_g}{F_{total}} \times \alpha \tag{6}$$

where $\alpha \geq 1$ is an equity adjustment factor that can overweight highly vulnerable regions.

3.4.2 Synthetic Minority Oversampling

For regions where new data collection is infeasible, we employ domain-informed data augmentation. Let $\mathcal{D}_g^{\text{minority}}$ denote underrepresented regions. We generate synthetic samples using:

$$x_{\text{synthetic}} = x_i + \lambda(x_j - x_i) + \epsilon \tag{7}$$

where $x_i, x_j \in \mathcal{D}_g^{\text{minority}}$, $\lambda \sim \text{Uniform}(0, 1)$, and $\epsilon \sim \mathcal{N}(0, \sigma^2 I)$ adds realistic noise.

Additionally, we apply environmental transformations that preserve geographic authenticity:

- Seasonal variations (vegetation indices, water levels)
- Weather conditions (rain intensity, cloud patterns)
- Topographic adjustments (terrain matching)

3.4.3 Domain Adaptation via Style Transfer

We employ CycleGAN-based domain adaptation [19] to transfer styles between well-represented and underrepresented regions while preserving flood semantics:

$$\mathcal{L}_{\text{cycle}} = \mathbb{E}_{x \sim \mathcal{D}_{\text{source}}}[||G_{\text{target}}(G_{\text{source}}(x)) - x||_1] \tag{8}$$

$$\mathcal{L}_{\text{identity}} = \mathbb{E}_{x \sim \mathcal{D}_{\text{target}}}[||G_{\text{target}}(x) - x||_1] \tag{9}$$

This allows us to create synthetic training samples with the environmental characteristics of underrepresented regions.

3.5 Phase 4: Equitable Model Development

3.5.1 Fairness-Constrained Training Objective

We propose a training objective that explicitly balances accuracy across geographic regions. Define the loss function:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{task}} + \lambda_{\text{fair}} \mathcal{L}_{\text{fairness}} \tag{10}$$

where:

$$\mathcal{L}_{\text{task}} = \mathbb{E}_{(x,y) \sim \mathcal{D}}[\text{BCE}(f(x), y)] \tag{11}$$

$$\mathcal{L}_{\text{fairness}} = \text{Var}_g[\text{IoU}_g] + \gamma \cdot \text{PDI} \tag{12}$$

BCE is binary cross-entropy, and $\text{Var}_g[\text{IoU}_g]$ penalizes variance in performance across regions. This encourages the model to perform consistently across all geographic contexts.

3.5.2 Multi-Task Learning with Regional Branches

For cases where shared representations fail to generalize, we propose a multi-task architecture:

$$f(x;g) = f^{(g)}_{\text{decoder}}(f_{\text{encoder}}(x)) \tag{13}$$

where f_{encoder} learns shared flood-relevant features and $f^{(g)}_{\text{decoder}}$ are region-specific decoder heads that adapt to local environmental characteristics.

3.5.3 Training Algorithm

Algorithm 2 Fairness-Constrained Model Training

Input: Augmented dataset $\mathcal{D}'$, regions $\mathcal{G}$, fairness weight λ_{fair}
Output: Trained model f^*

Initialize model f with ImageNet-pretrained encoder

repeat
  // Standard training step
  Sample minibatch $\mathcal{B} \sim \mathcal{D}'$
  Compute $\mathcal{L}_{\text{task}}$ on $\mathcal{B}$

  // Fairness regularization
  for each region $g \in \mathcal{G}$ **do**
    Sample validation batch $\mathcal{V}_g \sim \mathcal{T}_g$
    Compute IoU_g on $\mathcal{V}_g$
  end for
  Compute $\mathcal{L}_{\text{fairness}} = \text{Var}[\{\text{IoU}_g\}]$

  // Combined optimization
  $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{task}} + \lambda_{\text{fair}}\mathcal{L}_{\text{fairness}}$
  Update model parameters: $\theta \leftarrow \theta - \eta\nabla_\theta\mathcal{L}_{\text{total}}$
until convergence or max iterations

3.6 Justification for Proposed Methodology

3.6.1 Why Geographic Bias Quantification Works

The GRI metric is theoretically grounded in principles of proportional representation from fairness literature. By normalizing for area, flood frequency, and population exposure, GRI captures whether dataset coverage matches actual need rather than simply economic capacity to collect data. This addresses the root cause of geographic bias: opportunistic data collection that favors accessible, well-funded regions.

3.6.2 Why Cross-Regional Evaluation Reveals True Performance

Traditional evaluation using random train-test splits systematically underestimates performance disparities because test samples come from the same geographic distribution as training data. Stratified evaluation ensures that we test models on truly out-of-distribution regions, revealing performance gaps that would otherwise remain hidden until deployment. This is critical for disaster response applications where failures in underrepresented regions have severe consequences.

3.6.3 Why Bias-Aware Augmentation Improves Equity

Synthetic oversampling and domain adaptation address the fundamental constraint of limited data availability in underrepresented regions. By generating realistic synthetic samples that capture environmental characteristics of these regions, we provide the model with exposure to diverse geographic contexts during training. Research in domain adaptation [20] has shown that style transfer while preserving semantic content enables models to generalize across visual domains.

3.6.4 Why Fairness-Constrained Training Enhances Robustness

The fairness regularization term explicitly penalizes models that excel in some regions while failing in others. This forces the optimization process to find solutions that balance performance across geographic contexts rather than optimizing for average performance (which can be high even when some regions have very poor performance). Similar approaches have proven effective in algorithmic fairness literature for mitigating demographic disparities [21].

3.7 Baseline Approaches

To evaluate the effectiveness of our proposed methodology, we compare against four baseline approaches:

3.7.1 Baseline 1: Standard Training (Geographic Bias Ignored)

Description: Train state-of-the-art flood detection models (U-Net [22], DeepLabV3+ [23]) on existing datasets without any bias mitigation. Use standard random train-test splits.

Rationale: This represents current common practice in flood detection research and establishes the extent of geographic performance disparities under status quo conditions.

Difference from Proposed Approach: No geographic stratification in evaluation, no bias-aware augmentation, no fairness constraints in training objective.

3.7.2 Baseline 2: Simple Oversampling

Description: Duplicate samples from underrepresented regions to balance dataset sizes across regions, then train with standard loss function.

Rationale: This is a naive approach to address imbalance that does not add new information but ensures equal training exposure across regions.

Difference from Proposed Approach: Simple duplication vs. our domain-informed synthetic augmentation. No style transfer. No fairness-constrained objective.

3.7.3 Baseline 3: Transfer Learning Without Fairness Constraints

Description: Pre-train models on well-represented regions, then fine-tune on underrepresented regions using available data.

Rationale: Transfer learning is a standard approach for low-data scenarios and has shown promise in flood detection [13].

Difference from Proposed Approach: Sequential transfer learning vs. our joint fairness-constrained training. Does not explicitly optimize for cross-regional equity. May catastrophically forget well-represented regions during fine-tuning.

3.7.4 Baseline 4: Multi-Task Learning Without Fairness Constraints

Description: Train model with region-specific decoder heads (similar to our architecture) but using standard task loss only, without fairness regularization.

Rationale: Tests whether architectural choices alone (without fairness constraints) can address geographic bias.

Difference from Proposed Approach: Same architecture but lacks $\mathcal{L}_{\text{fairness}}$ term that explicitly penalizes performance variance across regions. Optimization does not explicitly seek equitable solutions.

3.8 Evaluation Protocol

All methods (proposed + baselines) will be evaluated using:

- Geographically stratified test sets ensuring equal representation
- Metrics: IoU, Precision, Recall, F1-score (per region and aggregate)
- Performance Disparity Index (PDI) and Gini coefficient
- Statistical significance testing (Kruskal-Wallis, Dunn's tests)

We hypothesize that our proposed approach will achieve:

1. Lower PDI than all baselines (more equitable performance)
2. Comparable or better aggregate performance than Baseline 1
3. Superior performance in underrepresented regions compared to all baselines
4. Minimal performance degradation in well-represented regions

3.9 Expected Outcomes

Based on the theoretical foundations and empirical evidence from related domains, we expect:

1. **Bias Quantification:** GRI analysis will reveal 3-10x overrepresentation of North America/Europe compared to Africa/South Asia, with SDM scores indicating strong correlation between GDP and dataset representation.

2. **Performance Disparities:** Baseline models will exhibit 10-20% lower IoU in underrepresented regions, with statistically significant differences (p ¡ 0.01) across geographic strata.

3. **Proposed Method Effectiveness:** Our fairness-constrained approach will reduce PDI by 40-60% compared to standard training while maintaining within 2-3% of baseline aggregate performance.

4. **Practical Impact:** Improved performance in underrepresented regions will translate to more reliable flood detection for 500M+ people in currently underserved areas.

4 Experimental Design

This section details the experimental setup for investigating geographic bias in flood detection systems, including dataset specifications, data collection procedures, preprocessing pipelines, and quality assurance protocols.

4.1 Datasets

Our experimental design utilizes a combination of existing publicly available datasets and newly collected data from underrepresented regions. This multi-source approach enables comprehensive bias analysis while addressing identified geographic gaps.

4.1.1 Primary Datasets

1. Sen1Floods11 Dataset

Description: Sen1Floods11 [8] is our primary benchmark dataset, containing 4,831 labeled image chips (512×512 pixels) from Sentinel-1 SAR imagery covering 11 major flood events globally.

Specifications:

- **Sensor:** Sentinel-1 C-band SAR (VV and VH polarizations)
- **Spatial Resolution:** 10 meters per pixel
- **Temporal Coverage:** 2018-2019
- **Geographic Distribution:** North America (45%), Europe (30%), Asia (20%), Other (5%)
- **Labels:** Binary flood masks (flooded/non-flooded) hand-labeled by experts
- **Format:** GeoTIFF files with corresponding label masks
- **Size:** Approximately 15 GB total

Use in Experiments:

- **Phase 1 (Bias Quantification):** Analyze geographic distribution, compute GRI scores for represented regions
- **Phase 2 (Baseline Evaluation):** Train baseline models, evaluate performance across geographic strata
- **Well-Represented Regions:** United States (Texas, Louisiana, Nebraska floods), Spain, Bosnia, Iran
- **Underrepresented Regions:** Limited coverage of Sub-Saharan Africa, South Asia, small island states

Access: Publicly available via IEEE DataPort (`https://ieee-dataport.org/open-access/sen1`

2. FloodNet Dataset

Description: FloodNet [9] provides high-resolution aerial imagery for post-disaster damage assessment from hurricanes Harvey and Florence.

Specifications:

- **Sensor:** UAV/aerial RGB imagery
- **Spatial Resolution:** 0.3-0.5 meters per pixel
- **Temporal Coverage:** 2017-2018
- **Geographic Coverage:** Texas and North Carolina (United States only)
- **Labels:** Multi-class semantic segmentation (building, road, water, tree, vehicle, pool, grass)
- **Images:** 2,343 high-resolution images
- **Size:** Approximately 8 GB

Use in Experiments:

- Demonstrate geographic bias in high-resolution damage assessment
- Baseline for urban flood detection in formal infrastructure contexts
- Contrast with informal settlement imagery from newly collected data

Limitations for Our Study: Extreme geographic concentration (single country, hurricane-only events) makes it useful primarily as an example of geographic bias rather than for training generalizable models.

Access: Publicly available via GitHub (`https://github.com/BinaLab/FloodNet-Challenge`)

3. Global Flood Database (GFD)

Description: Tellman et al. [10] compiled global flood observations from MODIS imagery spanning two decades.

Specifications:

- **Sensor:** MODIS Terra and Aqua satellites
- **Spatial Resolution:** 250 meters per pixel
- **Temporal Coverage:** 2000-2018 (913 large flood events)
- **Geographic Coverage:** Truly global, including underrepresented regions
- **Labels:** Flood extent polygons validated against news reports and disaster databases
- **Format:** Shapefiles and GeoTIFFs

Use in Experiments:

- Identify flood events in underrepresented regions for targeted data collection
- Validate geographic bias metrics (comparison of dataset coverage vs. actual flood occurrence)
- Generate balanced test sets by sampling events proportionally across regions

Access: Available via Cloud to Street (`https://www.cloudtostreet.com/`) and associated publications

4.1.2 Supplementary Satellite Imagery

Sentinel-1 SAR Archive

Description: European Space Agency's Sentinel-1 mission provides continuous global SAR coverage.

Specifications:

- **Revisit Time:** 6 days (with both satellites)
- **Coverage:** Global
- **Resolution:** 10-40 meters depending on mode
- **Availability:** Free and open access since 2014

Use in Experiments: Source new unlabeled imagery for underrepresented regions identified in Phase 1. We will download Sentinel-1 imagery for flood events documented in GFD but not present in Sen1Floods11, focusing on:

- Sub-Saharan Africa: Nigeria (2018, 2020), South Sudan (2019), Mozambique (2019)
- South Asia: Bangladesh (2017, 2019, 2020), Pakistan (2020), Nepal (2017)
- Southeast Asia: Myanmar (2018), Cambodia (2018), Vietnam (2020)
- Small Island States: Haiti (2018), Fiji (2018)

Access: Copernicus Open Access Hub (`https://scihub.copernicus.eu/`)

Sentinel-2 Optical Archive

Description: Optical imagery for visual validation and multi-modal experiments.

Specifications:

- **Revisit Time:** 5 days
- **Resolution:** 10-60 meters (13 spectral bands)
- **Coverage:** Global

Use in Experiments: Validate SAR-based flood detections, provide context for environmental characteristics, support multi-modal baseline experiments.

4.1.3 Newly Collected Data for Underrepresented Regions

Based on Phase 1 bias analysis, we will collect new labeled data for regions with GRI ¡ 0.5.

Target Regions and Events:

Region	Event	Date	Target Samples
Bangladesh	Monsoon flooding	July 2020	500 chips
Nigeria	Fluvial flooding	Sept 2020	400 chips
Mozambique	Cyclone Idai	March 2019	400 chips
Pakistan	Indus flooding	Aug 2020	400 chips
South Sudan	White Nile flooding	Oct 2019	300 chips
Haiti	Tropical storms	Oct 2018	300 chips
Myanmar	Monsoon flooding	Aug 2018	300 chips
Nepal	Monsoon flooding	July 2017	200 chips
Total			**2,800 chips**

Table 1: Planned new data collection from underrepresented regions

Data Collection Protocol:

1. Satellite Imagery Acquisition

- Download Sentinel-1 SAR imagery (VV and VH polarizations) for specified events
- Acquire pre-flood and post-flood image pairs (within 14 days of peak flooding)
- Extract 512×512 pixel chips matching Sen1Floods11 specifications
- Ensure spatial overlap with areas documented as flooded in GFD or news reports

2. Human Annotation Process

Annotator Recruitment:

- **Expert Annotators (Tier 1):** Recruit 3-5 remote sensing researchers with flood mapping experience through professional networks (IEEE, AGU, Remote Sensing journal contacts)
- **Regional Experts (Tier 2):** Partner with universities and NGOs in target regions (e.g., BRAC University in Bangladesh, University of Lagos in Nigeria) to recruit 5-10 annotators familiar with local geography and flood patterns
- **Compensation:** $15-25 USD per hour depending on expertise and region (adjusted for local cost of living)

Annotation Guidelines:

- Provide standardized annotation manual with examples from Sen1Floods11
- Include region-specific examples showing local environmental characteristics (tropical vegetation, urban patterns, terrain)
- Define clear decision rules for ambiguous cases (e.g., waterlogged fields vs. open water)
- Require annotators to complete 20-sample training set with feedback before production annotation

Annotation Tool: Web-based pixel-wise labeling interface (e.g., LabelBox, CVAT) allowing:

- Display of SAR imagery with adjustable contrast/brightness
- Side-by-side pre/post-flood comparison
- Reference Sentinel-2 optical imagery when available
- Polygon drawing tools for flood extent delineation

3. Inter-Rater Reliability

To ensure annotation quality, we implement a rigorous reliability assessment:

Overlap Sampling Design:

- 20% of all samples (560 chips) will be annotated by 3 independent annotators
- Stratified selection ensuring coverage across all target regions and flood severities
- Includes both obvious cases (clear flooding) and ambiguous cases (marginal inundation)

Reliability Metrics:

Pixel-wise Agreement:

$$\text{IoU}_{\text{inter}} = \frac{|A_1 \cap A_2 \cap A_3|}{|A_1 \cup A_2 \cup A_3|} \tag{14}$$

where A_i is the flood mask from annotator i. We require mean $\text{IoU}_{\text{inter}} \geq 0.75$ for acceptable reliability.

Fleiss' Kappa:

$$\kappa = \frac{\bar{P} - \bar{P}_e}{1 - \bar{P}_e} \tag{15}$$

where $\bar{P}$ is observed agreement and $\bar{P}_e$ is expected agreement by chance. We require $\kappa \geq 0.70$ (substantial agreement).

Quality Control Procedures:

- **Weekly calibration sessions:** Annotators discuss disagreements on challenging samples, refine guidelines
- **Expert adjudication:** For samples with $\text{IoU}_{\text{inter}} < 0.6$, senior researcher reviews and provides ground truth
- **Automated quality checks:** Flag annotations with suspiciously low/high flood percentages, unrealistic shapes
- **Continuous monitoring:** Track per-annotator agreement rates, provide feedback and retraining as needed

4. Validation Against Ground Truth

Where available, validate annotations against:

- News reports and disaster databases (EM-DAT, DFO)
- Social media geotagged flood imagery
- Government flood reports and emergency response documents
- Optical satellite imagery (Sentinel-2, Planet) for visual confirmation

Human Subjects Considerations:

IRB Status: This research does not involve direct interaction with affected populations or collection of personal data. Annotators are compensated professionals performing a technical task (image labeling). Therefore, IRB approval is not required under 45 CFR 46.

However, we follow ethical best practices:

- Informed consent from all annotators regarding data use and publication
- Fair compensation meeting or exceeding local professional standards
- No use of personally identifiable information from satellite imagery
- Sensitivity to potential trauma when annotating disaster imagery (provide mental health resources if needed)

4.2 Data Preprocessing

All datasets undergo standardized preprocessing to ensure consistency and quality for model training and evaluation.

4.2.1 Sentinel-1 SAR Preprocessing

SAR imagery requires specialized preprocessing to correct for sensor and terrain effects.

Processing Pipeline:

1. Radiometric Calibration

Convert digital numbers (DN) to sigma-naught backscatter coefficients:

$$\sigma^0 = \frac{\mathrm{DN}^2}{A_i^2} \tag{16}$$

where A_i is the absolute calibration constant from metadata.

Purpose: Ensures measurements are comparable across different acquisition dates and satellite orbits.

2. Speckle Filtering

Apply Lee Sigma filter with 7×7 window to reduce multiplicative speckle noise:

$$I_{\text{filtered}}(x, y) = \bar{I} + k \cdot (I(x, y) - \bar{I}) \tag{17}$$

where $\bar{I}$ is local mean, k is adaptive coefficient based on local variance.

Purpose: Reduce noise while preserving edges and flood boundaries.

Implementation: SNAP (Sentinel Application Platform) toolbox or Python with scikit-image

3. Terrain Correction (Range-Doppler)

Correct for geometric distortions using SRTM 30m DEM:

- Remove foreshortening and layover effects in mountainous terrain
- Project to geographic coordinates (WGS84)
- Resample to consistent 10m resolution

Purpose: Ensure geometric accuracy for multi-temporal comparison and geographic analysis.

4. Decibel Conversion

Convert backscatter to dB scale for improved dynamic range:

$$\sigma^0_{\text{dB}} = 10 \cdot \log_{10}(\sigma^0) \tag{18}$$

Purpose: Normalize values to approximately [-30, 10] dB range suitable for neural networks.

5. Normalization

Apply min-max normalization per polarization channel:

$$X_{\text{norm}} = \frac{X - \mu}{\sigma} \tag{19}$$

where μ and σ are computed from training set statistics (separately for VV and VH).

Purpose: Zero-mean, unit-variance inputs for stable neural network training.

6. Chip Extraction

Extract 512×512 pixel chips with 128-pixel overlap:

- Maintain spatial context for flood patterns
- Discard chips with ¿50% no-data values
- Store as two-channel arrays [VV, VH]

4.2.2 Sentinel-2 Optical Preprocessing

For multi-modal experiments and validation:

1. Atmospheric Correction

Apply Sen2Cor processor to convert Top-of-Atmosphere to Bottom-of-Atmosphere reflectance:

- Correct for atmospheric scattering and absorption
- Cloud/shadow masking using Scene Classification Layer

2. Band Selection

Select relevant bands for flood detection:

- Band 2 (Blue, 490nm): Water detection
- Band 3 (Green, 560nm): Vegetation differentiation

- Band 4 (Red, 665nm): Soil/water contrast
- Band 8 (NIR, 842nm): Water absorption, vegetation vigor
- Band 11 (SWIR1, 1610nm): Water detection, cloud penetration
- Band 12 (SWIR2, 2190nm): Flood/soil moisture

3. Normalization

Normalize each band to [0,1] based on typical reflectance ranges.

4.2.3 Label Preprocessing

1. Mask Standardization

Convert all flood masks to consistent binary format:

- 0 = non-flooded (including permanent water, land, no-data)
- 1 = flooded (temporary inundation)

For FloodNet's multi-class labels, extract water class and binarize.

2. Quality Filtering

Remove low-quality samples:

- Discard chips where annotators disagreed significantly ($\text{IoU}_{\text{inter}} < 0.6$)
- Remove chips with excessive cloud cover in validation imagery (¿30%)
- Exclude chips with sensor artifacts or processing errors

3. Class Balance Analysis

Compute flood pixel percentage per chip:

$$p_{\text{flood}} = \frac{\sum_{i,j} y_{ij}}{H \times W} \tag{20}$$

Categorize chips:

- No flood: $p_{\text{flood}} < 0.01$ (1%)
- Light flooding: $0.01 \leq p_{\text{flood}} < 0.1$
- Moderate flooding: $0.1 \leq p_{\text{flood}} < 0.5$
- Heavy flooding: $p_{\text{flood}} \geq 0.5$

Ensure balanced representation across categories in training/validation splits.

4.2.4 Geographic Stratification

Region Assignment:

Assign each sample to geographic strata based on coordinates:

Stratum	Definition
North America	USA, Canada, Mexico
Europe	EU countries, UK, Norway, Switzerland
East Asia	China, Japan, South Korea
South Asia	India, Bangladesh, Pakistan, Nepal, Sri Lanka
Southeast Asia	Myanmar, Thailand, Vietnam, Cambodia, Indonesia, Philippines
Sub-Saharan Africa	Nigeria, Kenya, South Sudan, Mozambique, others
Latin America	Brazil, Argentina, Colombia, others
Middle East	Iran, Iraq, Turkey, others
Small Island States	Haiti, Fiji, Pacific islands, Caribbean

Table 2: Geographic stratification for experimental design

Socioeconomic Metadata:

Attach GDP per capita (World Bank data) and Human Development Index (UNDP) to each region for bias analysis.

4.2.5 Data Splits

Training Set (60%):

- Base training: All Sen1Floods11 data (as-is, reflecting existing bias)
- Augmented training: Sen1Floods11 + newly collected data + synthetic samples

- Stratified by region to ensure representation in augmented version

Validation Set (20%):

- Geographically stratified: Equal samples per region (k=100-200 per stratum)
- Used for hyperparameter tuning and fairness-constraint calibration
- Includes samples from all flood severity categories

Test Set (20%):

- Geographically stratified: Equal samples per region (k=100-200 per stratum)
- Held-out until final evaluation
- Includes events temporally separated from training (2020-2021 events)

Split Methodology:

Algorithm 3 Geographically Stratified Data Splitting

Input: Dataset $\mathcal{D}$, regions $\mathcal{G}$, target split (60/20/20)
Output: $\mathcal{D}_{\text{train}}, \mathcal{D}_{\text{val}}, \mathcal{D}_{\text{test}}$

for each region $g \in \mathcal{G}$ **do**
  Extract samples $\mathcal{D}_g$ from region g
  Shuffle $\mathcal{D}_g$ randomly
  $n_g = |\mathcal{D}_g|$
  $\mathcal{D}_g^{\text{train}} \leftarrow \mathcal{D}_g[1 : 0.6 \cdot n_g]$
  $\mathcal{D}_g^{\text{val}} \leftarrow \mathcal{D}_g[0.6 \cdot n_g + 1 : 0.8 \cdot n_g]$
  $\mathcal{D}_g^{\text{test}} \leftarrow \mathcal{D}_g[0.8 \cdot n_g + 1 : n_g]$
end for

$\mathcal{D}_{\text{train}} = \bigcup_g \mathcal{D}_g^{\text{train}}$
$\mathcal{D}_{\text{val}} = \bigcup_g \mathcal{D}_g^{\text{val}}$
$\mathcal{D}_{\text{test}} = \bigcup_g \mathcal{D}_g^{\text{test}}$

4.2.6 Data Augmentation Implementation

Applied during training only (not to validation/test):

Standard Augmentations:

- Random horizontal/vertical flips (p=0.5)
- Random rotation (±15 degrees)
- Random brightness adjustment (±20%)
- Random noise addition ($\mathcal{N}(0, 0.05)$)

Domain-Specific Augmentations:

- Seasonal simulation: Adjust backscatter values to mimic different vegetation states
- Terrain variation: Apply local slope/aspect transformations
- Synthetic minority oversampling: As specified in Methodology (Section 3.3.2)

Implementation: PyTorch transforms with custom SAR-aware augmentation functions.

4.2.7 Computational Infrastructure

Storage:

- Raw data: 50 GB (Sentinel-1 archives)
- Preprocessed chips: 25 GB
- Augmented dataset: 100 GB (including synthetic samples)

Processing:

- Preprocessing: High-memory CPU nodes (64 GB RAM) for SNAP processing
- Training: NVIDIA A100 or V100 GPUs (40-80 GB VRAM)

- Estimated preprocessing time: 2-3 weeks
- Estimated annotation time: 4-6 weeks (with 10 annotators)

4.3 Summary of Experimental Design

Our experimental design combines:

- **3 primary datasets** with varying geographic coverage and resolution
- **2,800 newly labeled samples** from 8 underrepresented regions
- **Rigorous preprocessing** standardizing SAR and optical imagery
- **Inter-rater reliability** assessment ensuring annotation quality ($\kappa \geq 0.70$)
- **Geographic stratification** enabling unbiased performance evaluation
- **Multiple data splits** for baseline comparison and fairness-aware training

This comprehensive approach provides the foundation for rigorously testing our hypothesis that geographic bias in training data leads to performance disparities across regions, and that our proposed methodology can mitigate these disparities while maintaining overall model quality.

5 Evaluation

This section details the evaluation framework for assessing the effectiveness of our proposed fairness-constrained approach for geographic bias mitigation in flood detection systems. We specify performance metrics, experimental procedures, statistical testing methods, and expected outcomes.

5.1 Performance Metrics

We employ a comprehensive set of metrics to evaluate both technical performance and geographic equity.

5.1.1 Primary Technical Metrics

1. Intersection over Union (IoU)

The standard metric for semantic segmentation, measuring pixel-wise overlap between predicted and ground-truth flood masks:

$$\text{IoU} = \frac{|Y_{\text{pred}} \cap Y_{\text{true}}|}{|Y_{\text{pred}} \cup Y_{\text{true}}|} \tag{21}$$

where Y_{pred} is the predicted flood mask and Y_{true} is the ground truth.

Interpretation: IoU ranges from 0 (no overlap) to 1 (perfect overlap). Values above 0.7 are considered good for flood detection applications.

Justification: IoU is robust to class imbalance and directly measures the spatial accuracy of flood extent delineation, which is critical for disaster response decision-making.

2. Precision

The proportion of predicted flood pixels that are actually flooded:

$$\text{Precision} = \frac{TP}{TP + FP} \tag{22}$$

where TP = true positives (correctly identified flood pixels), FP = false positives (non-flood pixels incorrectly labeled as flood).

Interpretation: High precision means few false alarms. Critical for avoiding unnecessary evacuations or resource misallocation.

3. Recall (Sensitivity)

The proportion of actual flood pixels that are correctly detected:

$$\text{Recall} = \frac{TP}{TP + FN} \tag{23}$$

where FN = false negatives (flood pixels incorrectly labeled as non-flood).

Interpretation: High recall means few missed flood areas. Critical for ensuring all affected areas receive assistance.

4. F1-Score

The harmonic mean of precision and recall:

$$\text{F1} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{24}$$

Justification: Balances the trade-off between false alarms and missed detections, providing a single performance summary.

5. Accuracy

Overall pixel-wise classification accuracy:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{25}$$

where TN = true negatives (correctly identified non-flood pixels).

Note: While commonly reported, accuracy can be misleading due to class imbalance (most pixels are typically non-flooded). We report it for completeness but prioritize IoU and F1-score.

5.1.2 Fairness and Equity Metrics

These metrics quantify geographic performance disparities—our primary research focus.

1. Performance Disparity Index (PDI)

Measures the relative range of performance across geographic regions:

$$\mathrm{PDI} = \frac{\max_g \mathrm{IoU}_g - \min_g \mathrm{IoU}_g}{\mathrm{mean}_g \mathrm{IoU}_g} \tag{26}$$

where IoU_g is the mean IoU for region g.

Interpretation: PDI = 0 indicates perfect equity (all regions have identical performance). Higher PDI indicates greater geographic disparities. We target PDI ¡ 0.3 for acceptable equity.

Justification: PDI directly quantifies the geographic bias problem. Reduction in PDI demonstrates that our method achieves more equitable performance.

2. Gini Coefficient

Measures inequality in IoU distribution across regions:

$$\mathrm{Gini} = \frac{1}{2n^2 \bar{\mathrm{IoU}}} \sum_{i=1}^{n} \sum_{j=1}^{n} |\mathrm{IoU}_i - \mathrm{IoU}_j| \tag{27}$$

where n is the number of regions and $\bar{\mathrm{IoU}}$ is mean IoU.

Interpretation: Gini ranges from 0 (perfect equality) to 1 (maximum inequality). Lower is better. We target Gini ¡ 0.15.

Justification: The Gini coefficient is a well-established inequality metric from economics, providing complementary perspective to PDI.

3. Worst-Group Performance

The minimum IoU achieved across all geographic strata:

$$\mathrm{IoU}_{\text{worst}} = \min_g \mathrm{IoU}_g \tag{28}$$

Interpretation: Directly measures performance in the most underserved region. Critical for ensuring no population is left behind.

Justification: Fairness literature emphasizes that average performance can mask severe disparities. Worst-group performance ensures we focus on the most vulnerable populations.

4. Region-Specific Performance Gap

For each underrepresented region g_u, compare to best-represented region g_b:

$$\Delta_g = \text{IoU}_{g_b} - \text{IoU}_{g_u} \tag{29}$$

Interpretation: Performance gap in absolute IoU points. We target $\Delta_g < 0.10$ (10 percentage points) for all regions.

5.1.3 Stratified Metrics

We compute all metrics separately for:

- **Geographic strata**: 9 regions (North America, Europe, East Asia, South Asia, Southeast Asia, Sub-Saharan Africa, Latin America, Middle East, Small Island States)
- **Socioeconomic strata**: 4 quintiles by GDP per capita (Q1: lowest income, Q4: highest income)
- **Flood severity**: 4 categories (no flood, light, moderate, heavy)
- **Data representation**: Overrepresented (GRI ¿ 1.5), Balanced (0.7 $\leq$ GRI $\leq$ 1.5), Underrepresented (GRI ¡ 0.7)

This stratification enables fine-grained analysis of where and why performance disparities occur.

5.2 Experimental Procedure

5.2.1 Training Configurations

We train and evaluate 5 distinct model configurations:

Baseline 1: Standard Training (Geographic Bias Ignored)

- **Architecture:** U-Net with ResNet50 encoder (pretrained on ImageNet)
- **Training data:** Sen1Floods11 (as-is, reflecting existing bias)
- **Loss:** Binary cross-entropy (BCE)
- **Optimization:** Adam optimizer, lr=0.001, batch size=16
- **Training:** 100 epochs, early stopping (patience=10)
- **Purpose:** Represents current common practice baseline

Baseline 2: Simple Oversampling

- **Architecture:** Same U-Net as Baseline 1
- **Training data:** Sen1Floods11 + duplicate samples from underrepresented regions to balance dataset
- **Loss:** Binary cross-entropy
- **Optimization:** Same as Baseline 1
- **Purpose:** Tests whether naive balancing addresses bias

Baseline 3: Transfer Learning

- **Phase 1:** Pre-train on Sen1Floods11 (well-represented regions)
- **Phase 2:** Fine-tune on newly collected data (underrepresented regions)
- **Fine-tuning:** Lower learning rate (0.0001), freeze encoder initially
- **Purpose:** Standard transfer learning approach

Baseline 4: Multi-Task Learning (No Fairness Constraint)

- **Architecture:** Shared encoder + region-specific decoder heads
- **Training data:** Sen1Floods11 + newly collected data
- **Loss:** Weighted BCE across region-specific heads
- **Purpose:** Tests architectural solution without fairness regularization

Proposed Method: Fairness-Constrained Training

- **Architecture:** Same multi-task architecture as Baseline 4
- **Training data:** Sen1Floods11 + newly collected data + synthetic augmentation
- **Loss:** $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{task}} + \lambda_{\text{fair}}\mathcal{L}_{\text{fairness}}$ (from Methodology)
- **Fairness weight:** $\lambda_{\text{fair}} = 0.5$ (tuned on validation set)
- **Optimization:** Adam, lr=0.001, batch size=16
- **Purpose:** Our proposed approach combining architecture + fairness constraint + augmentation

5.2.2 Cross-Validation Strategy

Due to geographic stratification requirements, we employ **Stratified K-Fold Cross-Validation** (K=5) with geographic constraints:

Justification: This ensures that:

- Each region appears in both training and validation across folds
- Performance estimates are stable across different train/val splits
- We can quantify uncertainty in our measurements

Algorithm 4 Geographic Stratified K-Fold Cross-Validation

Input: Dataset $\mathcal{D}$, regions $\mathcal{G}$, folds K=5
Output: K performance estimates with regional stratification

for each region $g \in \mathcal{G}$ **do**
  Divide $\mathcal{D}_g$ into K equal folds: $\{\mathcal{D}_g^{(1)}, \ldots, \mathcal{D}_g^{(K)}\}$
end for

for $k = 1$ to K **do**
  // Construct training set for fold k
  $\mathcal{D}_{\text{train}}^{(k)} = \bigcup_{g \in \mathcal{G}} \bigcup_{j \neq k} \mathcal{D}_g^{(j)}$

  // Construct validation set for fold k
  $\mathcal{D}_{\text{val}}^{(k)} = \bigcup_{g \in \mathcal{G}} \mathcal{D}_g^{(k)}$

  // Train model on training set
  $M^{(k)} \leftarrow \text{Train}(\mathcal{D}_{\text{train}}^{(k)})$

  // Evaluate on validation set (per region)
  for each region $g \in \mathcal{G}$ **do**
    Compute $\text{IoU}_g^{(k)}$ on $\mathcal{D}_g^{(k)}$
  end for

  // Compute aggregate and fairness metrics
  Compute $\text{PDI}^{(k)}, \text{Gini}^{(k)}, \text{IoU}_{\text{worst}}^{(k)}$
end for

// Report mean and std across folds
Return: Mean and std of all metrics across K folds

5.2.3 Random Repeats

To account for training stochasticity (random initialization, data shuffling, augmentation), we perform **3 independent training runs** for each configuration with different random seeds.

Procedure:

- Set random seeds: 42, 123, 456
- For each seed, perform full 5-fold cross-validation
- Total: 5 configurations × 3 seeds × 5 folds = 75 trained models

Reporting: For each configuration, report mean ± standard deviation across 3 × 5 = 15 evaluations.

5.2.4 Held-Out Test Set Evaluation

After model selection via cross-validation, we perform final evaluation on the held-out test set (20% of data, geographically stratified, temporally separated).

Test Set Characteristics:

- Equal representation from all 9 geographic regions (k=150 samples each)
- Includes flood events from 2020-2021 (not seen during training)
- Balanced across flood severity categories
- Includes both Sen1Floods11 holdout and newly collected holdout samples

Evaluation Protocol:

- Load best model from each configuration (based on validation IoU)
- Run inference on entire test set
- Compute all metrics (technical + fairness) stratified by region, GDP, severity

- Generate confusion matrices, ROC curves, PR curves
- Visualize predicted flood maps vs. ground truth for qualitative assessment

5.3 Statistical Significance Testing

To rigorously determine whether our proposed method outperforms baselines, we employ multiple statistical tests.

5.3.1 Hypothesis Formulation

Null Hypotheses:

H0-1 (Technical Performance): The proposed method does not improve mean IoU compared to Baseline 1 (standard training).

H0-2 (Fairness): The proposed method does not reduce PDI compared to Baseline 1.

H0-3 (Worst-Group): The proposed method does not improve worst-group IoU compared to Baseline 1.

H0-4 (Regional Equity): There is no significant difference in IoU distributions across regions for the proposed method.

Alternative Hypotheses:

H1-1: Proposed method achieves higher mean IoU than Baseline 1.

H1-2: Proposed method achieves lower PDI than Baseline 1.

H1-3: Proposed method achieves higher worst-group IoU than Baseline 1.

H1-4: IoU distributions are more uniform across regions for the proposed method.

5.3.2 Test Selection and Justification

1. Paired t-Test (for mean comparisons)

Compare mean IoU between proposed method and each baseline:

$$t = \frac{\bar{d}}{s_d/\sqrt{n}} \tag{30}$$

where $\bar{d}$ is mean difference in IoU across the 15 runs (3 seeds × 5 folds), s_d is standard deviation of differences, $n = 15$.

Assumptions:

- Paired samples (same data splits across methods)
- Differences approximately normally distributed (verified via Shapiro-Wilk test)

Justification: Paired t-test accounts for correlation between measurements on the same data splits, providing more statistical power than independent t-test.

Significance threshold: $\alpha = 0.05$ with Bonferroni correction for 4 pairwise comparisons: $\alpha_{\text{adj}} = 0.0125$

2. Wilcoxon Signed-Rank Test (non-parametric alternative)

If normality assumption is violated, we use Wilcoxon signed-rank test:

$$W = \sum_{i=1}^{n} [\text{sgn}(d_i) \cdot R_i] \tag{31}$$

where R_i is the rank of $|d_i|$ and d_i is the difference in IoU for run i.

Justification: Distribution-free test, robust to outliers and non-normal differences.

3. Kruskal-Wallis H-Test (for regional equity)

Test whether IoU distributions differ significantly across geographic regions:

$$H = \frac{12}{N(N+1)} \sum_{g=1}^{G} \frac{R_g^2}{n_g} - 3(N+1) \tag{32}$$

where G is number of regions, R_g is sum of ranks for region g, n_g is sample size for region g, $N = \sum_g n_g$.

Null hypothesis: All regions have the same IoU distribution.

Application:

- For Baseline 1: We expect to reject H0 (significant regional differences exist)
- For Proposed method: We aim to *not* reject H0 (regional differences minimized)

Justification: Kruskal-Wallis extends the Wilcoxon test to multiple groups, testing our core hypothesis about geographic equity.

4. Dunn's Post-Hoc Test

If Kruskal-Wallis rejects H0, perform pairwise comparisons between regions with Bonferroni correction:

$$z_{ij} = \frac{\bar{R}_i - \bar{R}_j}{\sqrt{\frac{N(N+1)}{12} \cdot (\frac{1}{n_i} + \frac{1}{n_j})}} \tag{33}$$

Purpose: Identify which specific region pairs have significantly different performance.

5. Levene's Test (for variance homogeneity)

Test whether variance in IoU differs significantly across regions:

$$W = \frac{(N-G)}{(G-1)} \cdot \frac{\sum_{g=1}^{G} n_g(\bar{Z}_g - \bar{Z})^2}{\sum_{g=1}^{G} \sum_{i=1}^{n_g} (Z_{gi} - \bar{Z}_g)^2} \tag{34}$$

where $Z_{gi} = |\text{IoU}_{gi} - \tilde{\text{IoU}}_g|$ and $\tilde{\text{IoU}}_g$ is the median IoU for region g.

Purpose: Confirms that fairness-constrained training reduces not just mean differences but also variance across regions.

6. Bootstrap Confidence Intervals

For PDI and Gini coefficients, compute 95% confidence intervals via bootstrap (10,000 resamples):

- Resample the 15 runs (3 seeds × 5 folds) with replacement
- Compute PDI and Gini for each bootstrap sample
- Report 2.5th and 97.5th percentiles as confidence interval

Justification: PDI and Gini have complex distributions; bootstrap provides reliable confidence intervals without distributional assumptions.

5.3.3 Multiple Testing Correction

With multiple hypothesis tests, we control for false discovery rate using:

Bonferroni Correction: For m tests, use significance threshold α/m.

Application:

- 4 pairwise comparisons (Proposed vs. each Baseline): $\alpha_{\text{adj}} = 0.05/4 = 0.0125$
- Regional pairwise comparisons (9 regions = 36 pairs): $\alpha_{\text{adj}} = 0.05/36 = 0.0014$

Holm-Bonferroni: Step-down procedure, less conservative than Bonferroni, controls family-wise error rate.

5.3.4 Effect Size Reporting

In addition to p-values, we report effect sizes:

Cohen's d (for mean IoU differences):

$$d = \frac{\mu_{\text{proposed}} - \mu_{\text{baseline}}}{s_{\text{pooled}}} \tag{35}$$

Interpretation: $|d| < 0.2$ (small), $0.2 \leq |d| < 0.8$ (medium), $|d| \geq 0.8$ (large)

Justification: Effect sizes provide practical significance beyond statistical significance, indicating whether improvements are meaningful in real-world deployment.

5.4 Expected Results

Based on our methodology and related work findings, we present expected outcomes and how they support our research claims.

5.4.1 Primary Hypothesis: Geographic Bias Exists

Expected Result 1: Baseline 1 exhibits significant geographic performance disparities

Quantitative Predictions:

- North America IoU: 0.78 ± 0.04
- Europe IoU: 0.75 ± 0.05
- Sub-Saharan Africa IoU: 0.58 ± 0.08 (20 percentage points lower)
- South Asia IoU: 0.61 ± 0.07
- Performance Disparity Index: PDI $\approx$ 0.45-0.55
- Gini coefficient: 0.22-0.28
- Kruskal-Wallis test: p ¡ 0.001 (strong evidence of regional differences)

Support for Claims:

- Demonstrates that geographic bias in training data (Sen1Floods11) leads to measurable performance disparities
- Validates our problem statement that current systems perform poorly in underrepresented regions
- Provides baseline for measuring improvement

Visualization: Box plots showing IoU distributions by region will clearly show lower medians and higher variance for underrepresented regions.

5.4.2 Secondary Hypothesis: Simple Solutions Are Insufficient

Expected Result 2: Baselines 2-4 show modest improvements but cannot fully address bias

Baseline 2 (Simple Oversampling):

- Mean IoU: Similar to Baseline 1 (≈0.70)
- PDI: Reduced to 0.35-0.40 (modest improvement)
- Worst-group IoU: 0.62-0.65 (small improvement)
- Issue: Overfitting to duplicated samples, no new information added

Baseline 3 (Transfer Learning):

- Underrepresented regions IoU: 0.66-0.68 (improvement)
- Well-represented regions IoU: 0.70-0.72 (degradation due to catastrophic forgetting)
- PDI: 0.25-0.30 (better than Baseline 2 but still significant gaps)
- Issue: Sequential training doesn't jointly optimize for equity

Baseline 4 (Multi-Task, No Fairness):

- Mean IoU: 0.72 ± 0.03 (slight improvement)
- PDI: 0.30-0.35
- Issue: Architecture alone insufficient without explicit fairness objective

Support for Claims:

- Demonstrates that naive approaches provide incremental but incomplete solutions
- Justifies need for our comprehensive approach (architecture + augmentation + fairness constraint)

5.4.3 Primary Hypothesis: Proposed Method Achieves Equity

Expected Result 3: Fairness-constrained training significantly reduces geographic disparities

Quantitative Predictions:

- Mean IoU: 0.72 ± 0.03 (comparable to or better than baselines)
- North America IoU: 0.74 ± 0.04 (slight decrease acceptable)
- Sub-Saharan Africa IoU: 0.70 ± 0.05 (substantial improvement)
- South Asia IoU: 0.71 ± 0.04 (substantial improvement)
- Performance Disparity Index: PDI = 0.18-0.22 (60% reduction vs. Baseline 1)
- Gini coefficient: 0.10-0.12 (50% reduction)
- Worst-group IoU: 0.68-0.70 (17% improvement)
- Kruskal-Wallis test: p ¿ 0.05 (fail to reject H0, supporting regional equity)

Statistical Significance:

- Paired t-test vs. Baseline 1 for PDI: p ¡ 0.001, Cohen's d $\approx$ 1.2 (large effect)
- Bootstrap CI for PDI reduction: [0.20, 0.30] (does not include 0)
- Levene's test: p ¡ 0.05 for Baseline 1 (unequal variances), p ¿ 0.05 for Proposed (equal variances)

Support for Claims:

- Validates hypothesis that fairness-constrained training mitigates geographic bias
- Demonstrates practical significance (10+ percentage point improvement in underrepresented regions)
- Shows equity is achieved without sacrificing overall performance

5.4.4 Robustness Analysis

Expected Result 4: Proposed method maintains performance across flood severities

- Light flooding: IoU ≈ 0.68 (challenging due to ambiguity)
- Moderate flooding: IoU ≈ 0.74 (best performance)
- Heavy flooding: IoU ≈ 0.71 (good despite saturation effects)
- No significant interaction between region and severity (two-way ANOVA p ¿ 0.05)

Support for Claims: Demonstrates that equity improvements are not limited to specific flood types but generalize across severities.

Expected Result 5: Performance scales with data availability

- Ablation study varying newly collected data size (0%, 25%, 50%, 100%)
- PDI decreases monotonically as more underrepresented region data added
- Diminishing returns observed beyond 75% of target samples

Support for Claims: Informs practical data collection efforts—shows how much data is needed to achieve equity.

5.4.5 Qualitative Analysis

Expected Observations:

- **Visual inspection:** Baseline 1 misses flood extent in informal settlements and tropical vegetation (common in underrepresented regions). Proposed method correctly identifies these areas.
- **Error patterns:** Baseline 1 shows systematic false negatives in regions with unfamiliar environmental characteristics. Proposed method has more balanced error distribution.

- **Failure cases:** Both methods struggle with extreme cases (urban flash floods, mountainous terrain), but Proposed method fails more equitably across regions.

5.4.6 Addressing Alternative Explanations

Confound 1: Newly collected data quality

Control: Inter-rater reliability metrics (Fleiss' $\kappa \geq 0.70$) demonstrate annotation quality is comparable to Sen1Floods11.

Test: Train Baseline 1 on newly collected data only—if quality is issue, performance will be poor. We expect IoU ≈ 0.68-0.70, comparable to Sen1Floods11.

Confound 2: Model capacity

Control: Baseline 4 uses same architecture as Proposed method, isolating the effect of fairness constraint.

Expected: Baseline 4 PDI ≈ 0.30-0.35, Proposed PDI ≈ 0.18-0.22, confirming fairness regularization is key.

Confound 3: Data augmentation

Control: Ablation study with Proposed method but without synthetic augmentation.

Expected: PDI increases to 0.25-0.28, indicating augmentation contributes but fairness constraint is primary driver.

5.5 Summary of Evaluation Framework

Our comprehensive evaluation includes:

- **10 metrics** spanning technical performance and equity
- **5 model configurations** (1 proposed + 4 baselines)
- **5-fold geographic cross-validation** with 3 random repeats (75 models total)
- **6 statistical tests** with multiple testing correction

- **Stratified analysis** by region, GDP, severity, and data representation
- **Effect size reporting** for practical significance
- **Quantitative predictions** with expected ranges
- **Qualitative analysis** of failure modes and visual inspection

This rigorous framework will definitively establish whether our fairness-constrained approach successfully mitigates geographic bias while maintaining overall flood detection quality, advancing both the scientific understanding and practical deployment of equitable AI systems for disaster response.

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